LINEAR ALGEBRA · TOPIC 5 · LESSON 9 (MIT 18.06, LECTURE 9)

Independence, Basis and Dimension

Which funds are genuinely different? What is the smallest menu that still reaches everything? And how many independent directions does a menu really have? Three words answer it: independence, basis, dimension.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. **Conditions and important concepts**
   13 short definitions: independence, dependence, span, basis, dimension, bases of $C(A)$ and $N(A)$.
2. **Procedure**
   How to test independence, and how to find a basis and the dimension.
3. **2 × 2 examples**
   Nifty and bonds (independent); Nifty and its twin (dependent).
4. **3 × 3 examples**
   Nifty, bonds, midcap (a basis of $\mathbb{R}^3$); Nifty, bonds, balanced (dependent: bases of $C(A)$ and $N(A)$).
5. **Bigger example: five funds**
   Five funds, three genuinely different directions. With a fund picker.
6. **Connection: rank, dimension and n − r**
   Every fund is either a new direction or a redundancy.
7. **Final memory table**
   Every concept, its test and an example.

METHODS IN THIS LESSON: Independence | Independence test | Span | Basis | Dimension | Basis of C(A) | Basis of N(A)

PART 2 Prerequisites

Click any link to review it, then come back.

- Note 1: Ax = 0 (Lesson 7): R, pivots, rank, free variables, special solutions.

- The column space (Lesson 5) and the null space (Lesson 6).
- Arrow–Debreu building blocks (Lesson 3).

PART 3 Explanation

THE FUNDS USED IN THE EXAMPLES

Fund	Column: what ₹1 lakh does to (budget, return, exposure)
Nifty	$(1, 12, 1)$
Nifty ETF (a twin of Nifty)	$(1, 12, 1)$
Bonds	$(1, 7, 0)$
Balanced (half Nifty, half bonds)	$(1, 9.5, 0.5)$
Midcap	$(1, 15, 1.2)$

In this lesson each fund is a **vector** (its column). The questions are about the funds themselves: are they genuinely different, and how many do you really need?

1. Conditions and important concepts

1. What is linear independence?

DEFINITION · LINEARLY INDEPENDENT VECTORS

Vectors $v_1, v_2, \ldots, v_n$ are **independent** if $c_1 v_1 + c_2 v_2 + \cdots + c_n v_n = 0$ happens **only** when every c is 0.

In money terms:

Independent funds: no fund can be rebuilt from the others.

The only mix of them that changes nothing is "hold none of them".

2. What is dependence?

Vectors are **dependent** if some mix with not-all-zero c's gives 0. Example: Nifty, bonds, balanced:

$$-\tfrac{1}{2}\begin{bmatrix}1\\12\\1\end{bmatrix}-\tfrac{1}{2}\begin{bmatrix}1\\7\\0\end{bmatrix}+1\begin{bmatrix}1\\9.5\\0.5\end{bmatrix}=\begin{bmatrix}-0.5-0.5+1\\-6-3.5+9.5\\-0.5+0+0.5\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

That mix is a do-nothing trade, and it shows balanced $=\frac{1}{2}$ Nifty $+\frac{1}{2}$ bonds. So:

$$\boxed{\text{Dependent} = \text{a do-nothing trade exists} = \text{some fund is a mix of the others.}}$$

3. The independence test

Put the vectors as the columns of a matrix A. A mix $c_1v_1+\cdots+c_nv_n$ is just Ac, so the question "is there a non-zero c with $Ac=0$?" is the null space question from Note 1:

> **RULE · THE INDEPENDENCE TEST**
>
> The columns of A are independent $\iff N(A)=\{0\}\iff$ no free variables $\iff$ every column is a pivot column $\iff \text{rank}=n$.

4. The picture

Two vectors are dependent exactly when they lie on the same line (one is a multiple of the other): Nifty and its twin. **Three** vectors in $\mathbb{R}^3$ are dependent exactly when they lie in the same plane: Nifty, bonds and balanced all lie in the plane $R=7B+5E$ (Lesson 5).

5. Too many vectors are always dependent

More than m vectors in $\mathbb{R}^m$ are always dependent: $n>m$ means a free variable (Note 1). Example: any 4 funds measured by 3 rules.

$$\boxed{n>m \;\Rightarrow\; \text{dependent}}$$

6. What is span?

> **DEFINITION · SPAN**
>
> Vectors $v_1,\ldots,v_k$ **span** a space when the space consists of **all** their combinations. The columns of A span the column space $C(A)$.

In money terms: the span is every target the funds can reach. Nifty and bonds span the plane $R=7B+5E$.

7. What is a basis?

> **DEFINITION · BASIS**
>
> A **basis** for a space is a set of vectors that are (1) **independent** and (2) **span** the space.

$$\boxed{\text{Basis} = \text{reaches everything (span)} + \text{no waste (independent)}.}$$

In money terms: the smallest complete menu. Drop any fund and you lose something; add any fund and it's redundant.

8. The standard basis

The simplest basis of $\mathbb{R}^3$:

$$\begin{bmatrix}1\\0\\0\end{bmatrix}, \quad \begin{bmatrix}0\\1\\0\end{bmatrix}, \quad \begin{bmatrix}0\\0\\1\end{bmatrix}$$

Independent: $c_1(1,0,0) + c_2(0,1,0) + c_3(0,0,1) = (c_1, c_2, c_3)$, which is 0 only if every c is 0. Spanning: any (b_1, b_2, b_3) is b_1 of the first $+$ b_2 of the second $+$ b_3 of the third. In markets these are the **Arrow–Debreu securities** from Lesson 3: each pays 1 in exactly one scenario.

9. A space has many bases

Nifty, bonds, midcap are also a basis of $\mathbb{R}^3$ (3 pivots, worked below). In general:

$$\boxed{n \text{ vectors in } \mathbb{R}^n \text{ form a basis} \iff \text{their matrix is invertible} \iff n \text{ pivots}}$$

10. Dimension

> **DEFINITION · DIMENSION**
>
> Every basis of a space has the **same number** of vectors. That number is the **dimension** of the space.

$\mathbb{R}^3$ has dimension 3 (both bases above have 3 vectors). The plane $R = 7B + 5E$ has dimension 2. In money terms: the number of genuinely different directions you can trade.

11. Basis of the column space C(A)

> **METHOD · BASIS OF C(A) = THE PIVOT COLUMNS OF A**
>
> Reduce A to R, see which columns have pivots, then take **those columns of the original** A. They form a basis of $C(A)$, and $\dim C(A) = \text{rank}$.

> **IMPORTANT**
>
> Use the columns of A, **not** the columns of R. Row operations keep the null space but change the column space (Note 1). An example is in the 3 × 3 section.

12. Basis of the null space N(A)

> **METHOD · BASIS OF N(A) = THE SPECIAL SOLUTIONS**
>
> The special solutions (one free variable $= 1$, the others 0) are independent and span $N(A)$. So they are a basis, and $\dim N(A) = n - \text{rank}$.

Why independent: each special solution has a 1 in its own free slot where all the others have 0, so no mix of them can be zero unless every coefficient is.

13. Rank vs dimension

Matrices have a **rank**; spaces have a **dimension**. They meet here:

$$\boxed{\text{rank}(A) = \text{number of pivot columns} = \dim C(A)}$$

2. Procedure

A. Are these vectors independent?

Step	What to do
1	Put the vectors as the columns of A.
2	Row reduce $A \to R$.
3	Count the pivots.
4	Pivots $=$ number of vectors $\to$ **independent**. Fewer $\to$ **dependent**, and each special solution shows one dependency (which vector is a mix of which).

B. Basis and dimension of C(A) and N(A)

Step	What to do
1	Row reduce $A \to R$ and mark the pivot columns.
2	Basis of $C(A)$: the pivot columns **of** A. $\dim C(A) = r$.
3	Basis of $N(A)$: the special solutions. $\dim N(A) = n - r$.
4	Check: $\dim C(A) + \dim N(A) = r + (n - r) = n$.

3. 2 × 2 examples

Budget and return only.

Example A: independent (a basis of R²)

Nifty $(1, 12)$ and bonds $(1, 7)$:

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}$$

- Row 2 $-$ 12$\times$ row 1: $(0,\ 7 - 12 = -5)$.
- Row 2 $\div(-5)$: $(0, 1)$. Row 1 $-$ row 2: $(1, 0)$.

$$R = \begin{bmatrix} \boxed{1} & 0 \\ 0 & \boxed{1} \end{bmatrix} \qquad 2 \text{ pivots} = 2 \text{ vectors}$$

Therefore **independent**. Two independent vectors in $\mathbb{R}^2$ also span it, so Nifty and bonds are a **basis of** $\mathbb{R}^2$: every (budget, return) target is reachable in exactly one way. $\dim C(A) = 2$, $\dim N(A) = 0$.

Example B: dependent

Nifty $(1, 12)$ and its twin $(1, 12)$:

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} \xrightarrow{\text{row } 2 - 12 \times \text{row } 1} R = \begin{bmatrix} \boxed{1} & 1 \\ 0 & 0 \end{bmatrix} \qquad 1 \text{ pivot} < 2 \text{ vectors}$$

Therefore **dependent**. The special solution $(-1, 1)$ says $-1 \cdot \text{Nifty} + 1 \cdot \text{twin} = 0$: the twin is a copy. Both vectors lie on one line.

- Basis of $C(A)$: column 1 of A, $(1, 12)$. $\dim C(A) = 1$ (a line).
- Basis of $N(A)$: $(-1, 1)$. $\dim N(A) = 1$.
- Check: $1 + 1 = 2 = n$ ✓.

4. 3 × 3 examples

Example A: Nifty, bonds, midcap: a basis of R³

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix}$$

- A → U: row 2 $-\ 12\times$ row 1 $\to (0, -5, 3)$; row 3 $-$ row 1 $\to (0, -1, 0.2)$; row 3 $-\ 0.2\times$ row 2 $\to (0, 0, 0.2 - 0.6) = (0, 0, -0.4)$.
- U → R: row 3 $\div(-0.4) \to (0, 0, 1)$; row 2 $-\ 3\times$ row 3 $\to (0, -5, 0)$, then $\div(-5) \to (0, 1, 0)$; row 1 $-$ row 2 $-$ row 3 $\to (1, 0, 0)$.

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad 3 \text{ pivots} = 3 \text{ vectors}$$

Therefore **independent**, and three independent vectors in $\mathbb{R}^3$ span it: a **basis of** $\mathbb{R}^3$. Every (budget, return, exposure) target has exactly one portfolio, e.g. $(10, 111, 7.4) \to (5, 3, 2)$ (Lesson 8). $\dim C(A) = 3$, $\dim N(A) = 0$.

Example B: Nifty, bonds, balanced: dependent

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix}$$

- A → U: row 2 $-\ 12\times$ row 1 $\to (0, -5, -2.5)$; row 3 $-$ row 1 $\to (0, -1, -0.5)$; row 3 $-\ 0.2\times$ row 2 $\to (0, 0, 0)$.
- U → R: row 2 $\div(-5) \to (0, 1, 0.5)$; row 1 $-$ row 2 $\to (1, 0, 0.5)$.

$$R = \begin{bmatrix} \boxed{1} & 0 & 0.5 \\ 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 \end{bmatrix} \qquad 2 \text{ pivots} < 3 \text{ vectors}$$

Therefore **dependent**. The special solution $(-0.5, -0.5, 1)$ says balanced $= 0.5$ Nifty $+0.5$ bonds.

- **Basis of $C(A)$:** pivot columns 1 and 2 **of A:** Nifty $(1, 12, 1)$ and bonds $(1, 7, 0)$. $\dim C(A) = 2$: the plane $R = 7B + 5E$.
- **Basis of $N(A)$:** $(-0.5, -0.5, 1)$. $\dim N(A) = 1$.
- Check: $2 + 1 = 3 = n$ ✓.

WHY NOT THE COLUMNS OF R?

R's pivot columns are $(1, 0, 0)$ and $(0, 1, 0)$. They span the plane "third entry $= 0$". But $C(A)$ contains $(10, 105, 7)$ (₹7L Nifty + ₹3L bonds), whose third entry is 7. Different planes. Row operations changed the columns, so only A's own pivot columns span $C(A)$.

▼ **Check yourself (MIT problem 9.2): find a basis for the plane $x - 2y + 3z = 0$ in $\mathbb{R}^3$.**

The plane is $N\left(\begin{bmatrix} 1 & -2 & 3 \end{bmatrix}\right)$. $R = \begin{bmatrix} 1 & -2 & 3 \end{bmatrix}$ already: pivot x, free y and z. Special solutions: $y = 1$ gives $x = 2$, so $(2, 1, 0)$; $z = 1$ gives $x = -3$, so $(-3, 0, 1)$. That's a basis; the plane has dimension 2.

5. Bigger example: five funds

All five funds, three rules. Five vectors in $\mathbb{R}^3$, so (concept 5) they **must** be dependent. How many genuinely different directions are there?

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 & 15 \\ 1 & 1 & 0 & 0.5 & 1.2 \end{bmatrix} \quad \text{(Nifty, ETF, bonds, balanced, midcap)}$$

A → U:

- Row 2 $-$ 12$\times$ row 1: $0,\ 0,\ 7 - 12 = -5,\ 9.5 - 12 = -2.5,\ 15 - 12 = 3 \rightarrow (0, 0, -5, -2.5, 3)$.
- Row 3 $-$ 1$\times$ row 1: $0,\ 0,\ -1,\ -0.5,\ 1.2 - 1 = 0.2 \rightarrow (0, 0, -1, -0.5, 0.2)$.
- Column 2 (ETF): nothing left to pivot on. Skip.
- Row 3 $-$ 0.2$\times$ row 2: $0,\ 0,\ 0,\ -0.5 + 0.5 = 0,\ 0.2 - 0.6 = -0.4 \rightarrow (0, 0, 0, 0, -0.4)$.
- Column 4 (balanced): nothing left to pivot on. Skip. Column 5 (midcap): pivot -0.4.

U → R:

- Row 3 $\div(-0.4)$: $(0, 0, 0, 0, 1)$.
- Row 2 $-$ 3$\times$ row 3: $(0, 0, -5, -2.5, 0)$; then $\div(-5)$: $(0, 0, 1, 0.5, 0)$.
- Row 1 $-$ row 3 $-$ row 2: $1,\ 1,\ 1 - 1 = 0,\ 1 - 0.5 = 0.5,\ 1 - 1 = 0 \rightarrow (1, 1, 0, 0.5, 0)$.

$$R = \begin{bmatrix} \boxed{1} & 1 & 0 & 0.5 & 0 \\ 0 & 0 & \boxed{1} & 0.5 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

Pivot columns: 1, 3, 5 (Nifty, bonds, midcap). Free: 2, 4 (ETF, balanced). Rank $= 3$.

- **Independent?** No: 3 pivots $<$ 5 vectors.
- **Basis of $C(A)$:** Nifty $(1, 12, 1)$, bonds $(1, 7, 0)$, midcap $(1, 15, 1.2)$. Three independent vectors in $\mathbb{R}^3$, so $C(A) = \mathbb{R}^3$: $\dim C(A) = 3$.
- **Basis of $N(A)$:** special solutions $s_1 = (-1, 1, 0, 0, 0)$ (ETF $=$ Nifty) and $s_2 = (-0.5, 0, -0.5, 1, 0)$ (balanced $=$ half Nifty $+$ half bonds). $\dim N(A) = 5 - 3 = 2$.
- Check: $3 + 2 = 5 = n$ ✓.

Check s_2 in A: budget $-0.5 + 0 - 0.5 + 1 + 0 = 0$ ✓; return $-6 + 0 - 3.5 + 9.5 + 0 = 0$ ✓; exposure $-0.5 + 0 + 0 + 0.5 + 0 = 0$ ✓.

5 funds, but only 3 genuinely different directions.

Drop the ETF and balanced and you lose **nothing**: Nifty, bonds, midcap still reach every target. That smallest complete menu is a basis.

Try it: pick a menu

☑ Nifty ☑ ETF ☑ Bonds ☑ Balanced ☑ Midcap

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 & 15 \\ 1 & 1 & 0 & 0.5 & 1.2 \end{bmatrix} \rightarrow R = \begin{bmatrix} \boxed{1} & 1 & 0 & 0.5 & 0 \\ 0 & 0 & \boxed{1} & 0.5 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

Rank $r = 3$ with 5 funds: **dependent**.

Basis of $C(A)$ (pivot columns of A): Nifty, Bonds, Midcap. $\dim C(A) = 3$: every target in $\mathbb{R}^3$ is reachable.

Basis of $N(A)$ (special solutions): $s_1 = (-1, 1, 0, 0, 0)$ (ETF is a mix of the funds before it); $s_2 = (-0.5, 0, -0.5, 1, 0)$ (Balanced is a mix of the funds before it). $\dim N(A) = 2$.

Check: 3 + 2 = 5 funds.

6. Connection: rank, dimension and n − r

Every column of A is either a pivot column or a free column. So:

$$\underbrace{n}_{\text{funds}} = \underbrace{r}_{\dim C(A)\text{: new directions}} + \underbrace{(n-r)}_{\dim N(A)\text{: redundancies}}$$

Menu	n	$\dim C(A) = r$	$\dim N(A) = n - r$	Independent?
Nifty, bonds	2	2	0	yes
Nifty, twin	2	1	1	no
Nifty, bonds, midcap	3	3	0	yes (basis of $\mathbb{R}^3$)
Nifty, bonds, balanced	3	2	1	no
All five	5	3	2	no

How this ties the last lessons together:

Earlier idea	In this lesson's words
Column space $C(A)$ (Lesson 5)	the span of the columns; basis = pivot columns; dimension = rank
Null space $N(A)$ (Lesson 6)	basis = special solutions; dimension = $n - r$
Exactly one solution of $Ax = b$ (Lesson 8)	happens when the columns are a basis of the target space ($r = m = n$)
A do-nothing trade exists (Lesson 7)	the columns are dependent

READING IT AS A TRADER

$\dim C(A)$ is the number of **independent bets** the menu offers; adding funds beyond that adds no new exposure, only alternative routes. $\dim N(A)$ counts those alternative routes. A desk looking at 50 products driven by 3 risk factors has at most 3 independent bets, and at least 47 redundancies.

7. Final memory table

Concept	Meaning	Test / how to find it	Example
Independent	no vector is a mix of the others	rank $= n$ (no free variables), $N(A) = \{0\}$	Nifty, bonds, midcap
Dependent	some non-zero mix $= 0$	rank $< n$; special solutions show how	balanced $= \frac{1}{2}$ Nifty $+\frac{1}{2}$ bonds
Span	all combinations	columns of A span $C(A)$	Nifty, bonds span $R = 7B + 5E$
Basis	independent + spans	for $\mathbb{R}^n$: n pivots (invertible)	Nifty, bonds, midcap for $\mathbb{R}^3$
Dimension	vectors in any basis	count them	$\dim \mathbb{R}^3 = 3$
Basis of $C(A)$	reachable targets	pivot columns **of** A; $\dim = r$	five funds: Nifty, bonds, midcap
Basis of $N(A)$	do-nothing trades	special solutions; $\dim = n - r$	five funds: s_1, s_2
Key count		$n = r + (n - r)$	$5 = 3 + 2$